

SE Lab

(UE24CS341A)

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1. Architectural Choice

We selected a Microservices-oriented Architecture for the Blood Bank Emergency Matcher. The system is divided into focused services such as Emergency Request, Matching, Inventory, Donor, Notification, Audit & Reporting, and persistent data storage. This separation keeps responsibilities clear and allows individual services to evolve and scale independently.

2. Scenario-Specific Reasons

Reason 1 - Emergency scalability:

Emergency blood requests can arrive suddenly and require rapid processing. The Matching and Notification services can be scaled independently so that increased emergency traffic does not require scaling the entire system.

Reason 2 - Fault isolation and modularity:

Separating inventory, donor management, matching and notification reduces the impact of failures. For example, a temporary notification failure should not corrupt the blood inventory or emergency-request data. Each service also has a focused responsibility, making the system easier to maintain.

3. Security Advantage

Service boundaries provide a clear place to enforce authentication, authorization and controlled access to sensitive donor, patient and inventory information. Audit and reporting records can be maintained separately, supporting traceability of emergency requests, matching decisions and notifications.

4. Performance Benefit

The architecture supports independent scaling of high-demand operations. Matching can query inventory and donor information without making unrelated services part of the same processing unit, while notification can operate independently through its external messaging interface. This reduces bottlenecks and supports faster emergency response.

Conclusion:

The selected architecture provides a suitable balance of scalability, modularity, security and performance for an emergency blood matching system.